

Egor Chulkov

+7 (925) 114-07-20 | Egor.Chulkov@skoltech.ru | LinkedIn

Education

Skolkovo Institute of Science and Technology

2025 – Present

Ph.D. Student in Mathematics and Mechanics

Advisor: Prof. Oleg V. Vasilyev

Research Focus: Developing a complexity-driven pruning algorithm for implicit Constructive Solid Geometry (CSG) using interval arithmetic and localized Reverse Polish Notation (RPN) bit-masks.

- Achieved a massive reduction in memory footprint for highly complex scenes (e.g., 6.3 GB down to 16 MB) and orders of magnitude arithmetic speedup compared to unpruned baselines.
- Demonstrated significantly sparser spatial hierarchies and lower construction overhead against state-of-the-art Lipschitz and interval-based pruning methods.

Skolkovo Institute of Science and Technology

2023 – 2025

Master of Science in Mathematics and Computer Science

GPA: 5.0/5.0

Thesis: *Adaptive Constructive Solid Geometry with constant evaluation complexity for modeling complex implicitly defined objects*

- Designed the foundational adaptive spatial octree methodology and C++ CPU ray-tracer that served as the architecture for subsequent PhD-level pruning optimizations.

Moscow Institute of Physics and Technology

2019 – 2023

Bachelor of Science in Applied Mathematics and Physics

GPA: 4.5/5.0

Thesis: *Optimization of the characteristics of fiber optic distributed seismic sensors*

Publications & Preprints

- **Chulkov E.**, Vasilyev O. V., "Complexity-Driven Pruning for Implicit Constructive Solid Geometry." *Preprint*, 2026.
- **Chulkov E.**, Tikhotskiy S., Dubinya N., "Design of seismic sensors on DAS principle: analysis and numerical modeling," *GeoEurasia-2023*, Moscow.
- Tikhotskiy S., **Chulkov E.**, "On the possibility to design fiber optic seismic with the prescribed angular sensitivity diagram," *BalticPetroModel 2022*, St. Petersburg.

Patents

- **Chulkov E.**, Vasilyev O. V., "Computer-Aided Design System for Modeling Complex Implicitly Defined Objects." *Patent Application RU 2025131729*, Filed Nov 2025. **(Patent Pending)**
- Nachev V., Novikova E., Vasilev I., **Chulkov E.**, Markov R., Sokolov V., Khudykin A., "Method and System for Early Detection and Prediction of Fire Spread in Forest Areas." *Patent RU 2 849 054 C1 (MIPT)*, Published Oct 2025.

Professional Experience

TESIS

Software Engineering Intern (Thermal Radiation Spectral Model)

Summer 2024

Moscow, Russia

- Engineered a C++ solver for computing equilibrium temperatures in convex enclosures.
- Implemented VRML file parsing functionality for geometry import.
- Implemented an analytical method based on Narayanaswamy's algorithm, improving accuracy up to 99%.

Sakhalin Energy

2023 – 2024

Research Engineer (Geomechanics Research Project)

Moscow, Russia

- Developed a novel mathematical algorithm for calculating pore pressure field dynamics in producing wells.
- Engineered a calculation module to determine sand occurrence intensity based on well operation modes.
- Implemented inverse Laplace transform using Gaver-Stehfest algorithm for pressure field analysis.

Moscow Institute of Physics and Technology

2022 – 2024

Research Engineer, Laboratory of Wellbore Exploration

Moscow, Russia

- Conducted fiber deformation modeling using ABAQUS finite element analysis software.
- Performed analytical studies on spirally twisted optical fiber deformation under plane wave influence.

Technical Skills

- **Computer Graphics:** F-rep, SDF, SVO, CSG, R-functions, OpenVDB
- **Scientific Computing:** C++, CUDA, OpenMPI, Python (NumPy, SciPy, Pandas), MATLAB
- **Simulation & Analysis:** ABAQUS, Wavelet Transform, Gaver-Stehfest Algorithm
- **Development Tools:** Git, Docker, Django
- **Languages:** English (Upper-Intermediate), Russian (Native), French (Elementary)

Honors and Awards

- Academic Excellence Award, Skolkovo Institute of Science and Technology (2025)
- Prize winner of Phystech Olympiad (2018) and Phystech International Olympiad (2019)